

6. Branch Protection, Conductor, Converter, Loss, and Thermal Calculations

6.1 Branch Fuse Protection Schedule

To prevent thermal overload or fire hazard in the external cabling and terminals, fast-acting glass fuses ($5 \times 20\text{mm}$) are placed on each positive output rail downstream of the ATX breakout board. Fuse ratings are calculated to be below the maximum current capacity of 18AWG hook-up wire (10A) and within the continuous rating of the PLENTY COMPUTER (ATX500WS) source PSU.

6.2 Conductor Ampacity & Voltage Drop Calculations

Standard hook-up wire used for internal high-current connections is 18AWG stranded copper

$$R_{total} = R_{18AWG} \times L = 0.02095 \, \Omega/\text{m} \times 0.4 \, \text{m} = 0.00838 \, \Omega$$

$$\Delta V = I_{max} \times R_{total} = 5 \, \text{A} \times 0.00838 \, \Omega = 0.0419 \, \text{V} \, (41.9 \, \text{mV})$$

$$\text{Percentage Voltage Drop} = \left(\frac{0.0419\text{V}}{5.0\text{V}} \right) \times 100\% = 0.84\%$$

Verification Result: The calculated voltage drop of 0.84% is well below the maximum allowable lab limit of 3.0%, ensuring tight load regulation.

6.3 Converter Efficiency & Power Loss Analysis

The adjustable output sub-system utilizes an XL6009 / LM2596 Buck-Boost Converter Module powered from the protected + 12V rail.

Power Output (P_{out}) & Power Input (P_{in}):

$$P_{out} = V_{out} \times I_{out} = 5.0V \times 2.0A = 10.0 \text{ W}$$

$$P_{in} = V_{in} \times I_{in} = 12.0V \times 0.98A = 11.76 \text{ W}$$

Converter Efficiency (η) & Internal Power Dissipation (P_{loss}):

$$\eta = \left(\frac{P_{out}}{P_{in}} \right) \times 100\% = \left(\frac{10.0W}{11.76W} \right) \times 100\% = 85.03\%$$

$$P_{loss} = P_{in} - P_{out} = 11.76W - 10.0W = 1.76 \text{ W}$$

6.4 Thermal Rise & Heatsink Calculations

Estimated Temperature Rise (ΔT) & Operating Temperature ($T_{operating}$):

$$\Delta T = P_{loss} \times \theta_{JA} = 1.76\text{W} \times 25^\circ\text{C/W} = 44.0^\circ\text{C}$$

$$T_{operating} = T_{ambient} + \Delta T = 25.0^\circ\text{C} + 44.0^\circ\text{C} = 69.0^\circ\text{C}$$

Thermal Safety Verification: The projected silicon junction/heatsink temperature of 69.0c is significantly lower than the IC thermal shutdown threshold confirming safe continuous operation within the sealed acrylic enclosure.